Instructor: Prof. Amini
Date: 1404/___/___

Sharif University
of Technology

Advanced Traffic (20–551)
Class Project Part 2

Part 2: Object-Oriented Programming (OOP) for Traffic Elements

Goals:

1. Apply object-oriented programming (OOP).
2. Define network objects.
3. Define agents and their capabilities.

Tasks

1. Implement classes (e.g., `Lane` and `Intersection`).
2. Each class should expose relevant functionality (e.g., traffic signal control) and be able to store state/data.
3. Vehicles should travel along lanes and interact at intersections.

Outputs

1. One or more Python files implementing the features described above.
2. A brief report about the code (inline documentation in all files and a `README.md` file).
3. Present the code and answer questions.

Time

Weeks 1–2

Grade Weight

5%. In this part, you cannot fully evaluate whether the code is effective. Most of the points for this part will be combined with other parts. The current score evaluates only the structure of your classes and how well they are designed for future extension.